Learning Objectives 8.1, 8.2: I can articulate the significance of the displacement current and calculate it, given a time-varying electric field. I can articulate the meaning of and calculate the relaxation time for a variety of materials.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A coil in a changing field.** A flat square coil of $N = 50$ turns, 10 cm on a side, lies in the xy plane in a uniform field $\mathbf{B} = \hat{\mathbf{z}}\, B_0 \cos \omega t$ with $B_0 = 50$ mT at 60 Hz. We traverse the coil counterclockwise as seen from $+z$.

   (a) Find the flux through one turn and the amplitude of the coil's EMF.

   Φ amplitude =

   EMF amplitude =

   (b) At an instant when $\mathbf{B}$ points along $+\hat{\mathbf{z}}$ and is growing, which way does the induced current circulate, seen from $+z$? Explain with Lenz's law and with the sign of the EMF.

   (c) Starting from the integral form of Faraday's law for a path that does not move, derive $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$.

2. **A moving bar and a resistor.** A bar of length $\ell = 0.3$ m slides at $v = 4$ m/s along two rails in a uniform field $B = 0.5$ T into the page. A $2\,\Omega$ resistor closes the rails, and the bar and rails have no resistance.

   (a) Find the EMF and the current.

   EMF = ____

   I = ____

   (b) The magnetic force $I\boldsymbol{\ell} \times \mathbf{B}$ of Lesson 28 acts on the current in the bar. Find its magnitude and say which way it points.

   F = ____

   (c) Show that the mechanical power needed to keep the bar moving equals the power the resistor dissipates.

3. **Displacement current in a capacitor.** A parallel-plate capacitor has plates of area $A = 2$ cm^2 separated by $d = 0.5$ mm of a dielectric with $\epsilon_r = 4$. A 100 MHz source applies $v = 5\cos\omega t$ V. Ignore fringing.

   (a) Find the amplitude of the displacement current density between the plates.

   J_d amplitude =

   (b) Find the total displacement current between the plates, and compare it with the current $C\,dv/dt$ in the wire.

   i_d amplitude =

   (c) Explain, using two surfaces that share one path around the wire, why Ampere's law needs the displacement current to give a single answer here.

4. **Conduction or displacement current?** Wet soil has $\sigma = 10^{-2}$ S/m and $\epsilon_r = 10$.

   (a) For a field $E_0 \cos \omega t$, derive the ratio of the amplitudes of the conduction and displacement current densities.

   (b) Find the ratio at 1 MHz and at 1 GHz, and the frequency at which the two currents are equal.

   1 MHz: ____

   1 GHz: ____

   equal at ____

   (c) Is the soil a good conductor or a good insulator at each frequency?

5. **Relaxation time.** A uniform material has $\sigma = 10^{-4}$ S/m and $\epsilon_r = 3$. At $t = 0$ a free charge density ρ_0 sits inside it.

   (a) Starting from the continuity equation, derive $\rho_v(t)$.

   (b) Find the relaxation time τ_r, and the time for the charge inside to fall to 1% of ρ_0.

   τ_r = ____

   $t_{1\%}$ = ____

   (c) Where does the charge go?

Documentation: _____